Hospital information and patient Management system

1. Objective

The objective of the Hospital Information and Patient Management System is to develop a centralized web application using Python, Django REST Framework, and MySQL to manage patient records, doctor details, appointments, medical histories, prescriptions, billing, and administrative activities. The system aims to simplify registration and appointment scheduling, secure patient information, reduce paperwork and manual errors, improve departmental coordination, and provide authorized users with quick access to accurate records.

2. Existing System

In the existing system, hospital activities are often managed through paper records, registers, spreadsheets, or separate applications. Patient details, doctor schedules, appointments, medical reports, and billing information may be maintained independently by different departments.

Disadvantages:

- Time-consuming patient registration and record retrieval.
- Risk of lost, damaged, or duplicated records.
- Appointment scheduling conflicts and billing errors.
- Difficulty sharing information between departments.
- Increased paperwork and administrative workload.
- Limited access control, backup, and reporting capabilities.

3. Proposed System

The proposed system is a centralized web application that automates hospital operations through Django REST Framework APIs and a MySQL database. It enables authorized users to manage patients, doctors, appointments, medical records, prescriptions, billing, and reports through a common platform.

Advantages:

- Centralized and secure storage of hospital information.
- Faster patient registration and record retrieval.
- Efficient appointment booking and scheduling.
- Easy access to authorized medical histories and prescriptions.
- Improved billing accuracy and payment tracking.
- Role-based authentication and access control.
- Reduced paperwork and improved departmental coordination.
- Simplified report generation, data backup, and record maintenance.

Roles and Responsibilities – Hospital Information and Patient Management System

S. No.	Role	Responsibilities
1	Administrator	Manage user accounts, assign roles and permissions, manage hospital departments, configure system settings, and monitor system activities.
2	Receptionist	Register new patients, update patient details, schedule appointments, manage appointment status, and maintain registration records.
3	Doctor	View assigned appointments, access patient medical history, record diagnoses, prescribe medicines, and update consultation details.
4	Nurse	Record patient vital signs, maintain nursing observations, assist doctors during treatment, and update patient-care records.
5	Billing Staff	Generate patient invoices, calculate consultation and treatment charges, record payments, and maintain billing reports.
6	Patient	View personal information, check appointment details, and access permitted medical reports, prescriptions, and billing information.
7	Pharmacist	Manage medicine inventory, verify prescriptions, dispense medicines, and maintain stock records, if pharmacy management is included.
8	Laboratory Technician	Manage laboratory test requests, record test results, and make authorized reports available to doctors.

User Access Control Table

S. No.	Role	Patient Management	Doctor Management	Appointment Management	Medical Records	Billing	User Management
1	Administrator	Full Access	Full Access	Full Access	Authorized Administrative Access	View Reports	Full Access
2	Receptionist	Create, View, Update	View	Create, View, Update, Cancel	Limited Access	View Billing Status	No Access
3	Doctor	View Assigned Patients	View Own Profile	View Assigned Appointments	Create, View, Update Clinical Records	View Relevant Bills	No Access

4	Nurse	View Assigned Patients	View Doctor Details	View Assigned Appointments	Record Vital Signs and Nursing Notes	No Access	No Access
5	Billing Staff	View Basic Patient Details	View Doctor Details	View Appointment Details	No Access	Create Bills, View and Update Payments	No Access
6	Patient	View Own Details	View Available Doctor Details	Book and View Own Appointments	View Permitted Own Records	View Own Bills	Manage Own Account
7	Pharmacist	View Necessary Patient Details	View Prescription Information	No Access	View Prescriptions Only	No Access	No Access
8	Laboratory Technician	View Necessary Patient Details	View Test Requests	No Access	Record and Update Lab Results	No Access	No Access

Access control notes:

- Full Access: Create, view, update, and delete, where permitted by policy.
- Limited Access: Access only to the information necessary for the assigned role.
- No Access: The role cannot access that module.
- Medical records should be restricted to authorized healthcare staff and patients viewing their own permitted records.

Technology Stack – Hospital Information and Patient Management System

S. No.	Technology	Purpose
1	Python	Implements the backend logic and hospital management functionality.
2	Django	Handles application structure, database models, authentication, routing, and business logic.
3	Django REST Framework	Develops REST APIs to create, retrieve, update, and delete hospital records.
4	MySQL	Stores patient details, doctor information, appointments, medical records, prescriptions, and billing data.

5	HTML5	Structures the web pages and forms.
6	CSS3	Styles the user interface, including layouts, colors, and fonts.
7	JavaScript	Adds client-side interactivity, form validation, and dynamic page behavior.
8	Bootstrap 5	Creates responsive pages that work across desktops, tablets, and mobile devices.
9	Django ORM	Performs database operations using Python models instead of writing SQL queries for every operation.
10	DRF Serializers	Validates incoming API data and converts Django model objects into JSON-compatible representations.
11	Django REST Framework Permissions	Restricts API operations according to user roles, such as administrator, doctor, receptionist, and patient.
12	Postman	Tests API endpoints using GET, POST, PUT, PATCH, and DELETE requests.
13	Visual Studio Code	Provides an environment for writing, editing, and debugging the application code.
15	Git and GitHub	Tracks code changes, maintains version history, and supports source code collaboration.
16	Web Browser	Allows users to access the hospital management application.